HOLLOW VAULT

AI FSM One-Page Design · Assignment 3: Systemic Games — Autonomous Agents

CRYPT SENTINEL + THE SHADE

Top-Down Action-Adventure Roguelike

GAME CONCEPT

Hollow Vault is a top-down genre action-adventure rogue-like in an underground cursed treasury. The character (a relic thief) has to work his way through procedurally obtained vault rooms, loot relics and get out alive. The vault is guarded by two AI characters: the Crypt Sentinel (heavy armoured undead) and the Shade (a ghostly sound reactive stalker). The two agents relate to one another as well as to the player, giving rise to overlapping menace dynamics.

① CRYPT SENTINEL

PROPERTIES

Type: Undead Armoured Guardian
Speed: 1.8 m/s patrol / 3.2 m/s charge
HP: High — tank archetype
Sensor: Vision cone 60° / 9 m range
Weapon: Cursed Greatsword (melee)
Memory: Tracks last seen player pos. 4s

Crypt Sentinel

STATES

PATROL	Waypoint loop. Vision cone active.
INVESTIGATE	Movements to last known noise/sight pos [FB].
COMBAT	Charges & attacks player at melee range.
STAGGERED	Hit by player — brief stun. [FB]
RALLYING CRY	Summons Shade if within 10m. [FB]
RETURN	Returns to patrol after losing player.

② THE SHADE

PROPERTIES

Type: Spectral Stalker (intangible)
Speed: 2.5 m/s drift / 4.5 m/s hunt
HP: Low — avoids direct combat
Sensor: Sound radius 8m + light 5m
Attack: Stamina drain on contact
Weakness: Torch light — forced retreat

The Shade

STATES

DRIFT	Floats along room perimeter slowly.
ALERT	Audible noise-- orients to source. [FB]
STALK	Silently moves toward sound origin.
HUNT	Locks onto player. Closes fast.
RETREAT	Exposed to torch -- flees to shadow.
RALLIED	Called by Sentinel — hunts player.

OBJECT RELATIONSHIPS

Player → Sentinel
Entering vision cone triggers INVESTIGATE. Hitting Sentinel triggers STAGGERED. Fleeing causes RETURN.

Player → Shade
Footsteps/actions produce sound. Equipping torch forces RETREAT. Contact drains stamina.

Sentinel → Shade
RALLYING CRY pings Shade within 10m. Shade enters RALLIED state and hunts player.

Shade → Sentinel
Shade stalking near Sentinel shares player position (memory event).

Sentinel → Player
Greatsword attack in COMBAT deals damage. STAGGERED gives player an opening window.

Shade → Player
HUNT contact drains stamina. Slower player makes more noise — can alert Sentinel.

PERCEPTION / SENSORS

Sentinel — Vision Cone
60° FOV, 9m. Raycasted through walls. Active in PATROL/COMBAT. Memory: 4s after sight lost.

Shade — Sound Radius
8m sphere. Triggered by footsteps, attacks, opening chests, landing from jumps.

Shade — Light Sensor
5m radius. Detects lit torch. Immediately triggers RETREAT if active.

FEEDBACK STATES

! **SENTINEL SPOTS PLAYER**
'!' icon above head + metallic armour clang SFX.

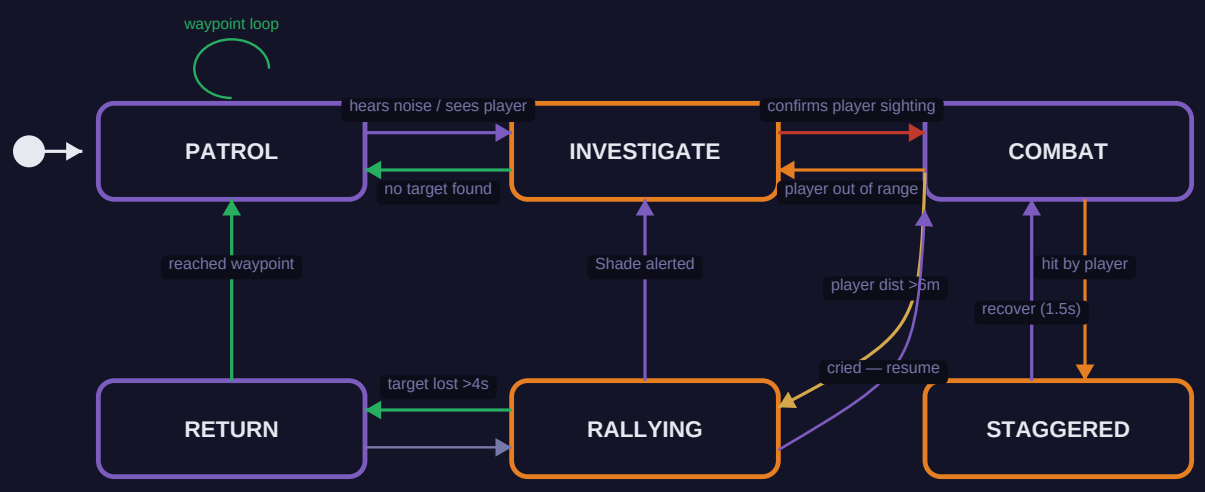
? **SENTINEL INVESTIGATES**
'?' icon above head + slow heavy footstep SFX.

~ **SHADE ALERT**
Screen-edge ripple VFX + low resonant hum SFX.

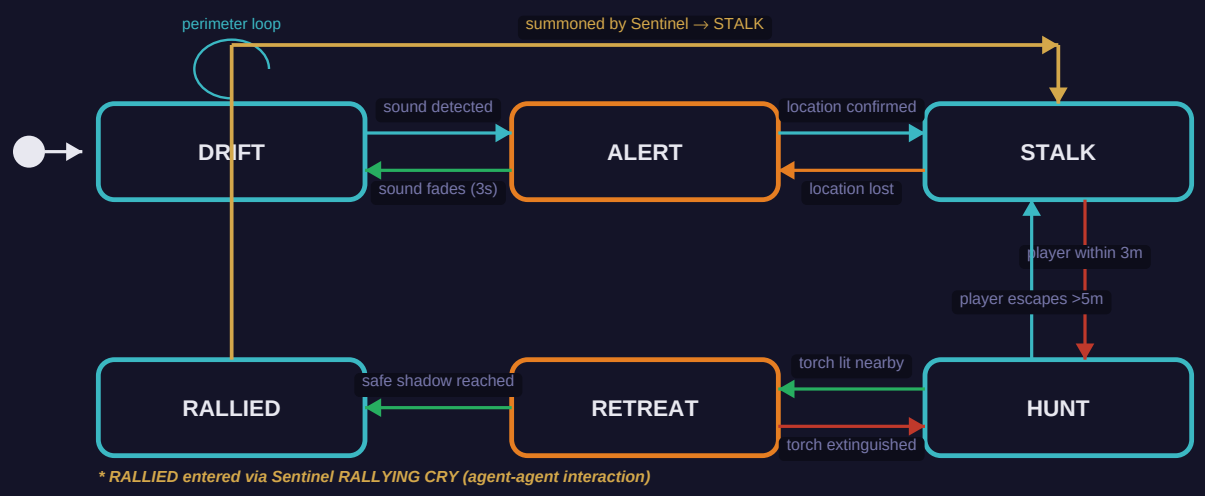
♦ **SHADE HUNTS**
Heartbeat SFX loop + darkened screen vignette.

FSM DIAGRAMS — UML STATE NOTATION

① CRYPT SENTINEL — Finite State Machine



② THE SHADE — Finite State Machine



** RALLIED entered via Sentinel RALLYING CRY (agent-agent interaction)*

GAMEPLAY SCENARIOS + DESIGN RATIONALE

GAMEPLAY SCENARIOS

SCENARIO A — The Chest Ambush

1. Player enters treasure room. Crypt Sentinel is in mid-PATROL, left vision cone sweep.
2. Player opens a chest — generates noise event. Shade (in adjacent hall) enters ALERT. Screen-edge ripple VFX appears.
3. Shade transitions ALERT → STALK. Silently drifts toward chest room. Player is unaware.
4. Player steps within 3m of Shade — Shade enters HUNT. Heartbeat SFX rises. Screen vignette darkens.
5. Shade contact drains stamina. Player slows — footsteps become louder noise events.
6. Louder footsteps exceed Sentinel sound threshold. Sentinel enters INVESTIGATE. '?' icon appears.
7. Sentinel gets line-of-sight. Transitions INVESTIGATE → COMBAT. '!' + clang SFX. Charges player.
8. Player lights torch — Shade immediately enters RETREAT. Sentinel is now the only threat.
9. Player escapes corridor. Sentinel loses sight >4s → RETURN. Shade reaches shadow → DRIFT. Tension resolves.

SCENARIO B — The Sentinel's Call

1. Player sneaks (no footsteps). Sentinel on PATROL. Shade DRIFT. No aggro.
2. Player misjudges Sentinel's cone — detected. Sentinel: PATROL → INVESTIGATE → COMBAT.
3. Player hits Sentinel — COMBAT → STAGGERED. Brief window to flee.
4. Sentinel HP threshold reached. Triggers RALLYING CRY. Calls Shade within 10m.
5. Shade enters RALLIED → immediately transitions to STALK → HUNT toward player.
6. Player flanked: Sentinel recovering behind, Shade closing from the side.
7. Player ducks into shadow alcove — Shade enters RETREAT. One threat neutralized.
8. Player kites Sentinel into narrow corridor. Sentinel loses sight → RETURN. Escape.

DESIGN RATIONALE

Objects, Properties & Behaviours (Fullerton)

Every AI is an individual object with properties that have direct generative behaviours. The high HP and vision cone of the Sentinel turn it into a constant danger of the presence - infers slowly, but it will come back. Low HP and sound/light sensors of The Shade render the latter an attrition threat - weak, but it will penalize loud play. After Fullerton, behaviours (states) are developed out of properties: a fragile agent cannot confront his opponent head-on, thus RETREAT is a property which the Shade needs to have.

Player-Agent Relationships

The two agents also limit the various players behaviours. The Sentinel is a penalty of showing up (getting into its cone). The Shade penalizes noises and rewards the possession of a torch. The part of the battle that is the most dramatic is the decision to use the torch: making it will scare away the Shade but make the Sentinel see more clearly. Both aspects are under the player control of building meaningful moment-to-moment decisions.

Agent-Agent Relationships

Intentional co-dependence is created by the RALLYING CRY mechanic. There is no one of the agents who cannot be beaten. The Shade fills the gaps in the vision cone of the Sentinel, and the Shade retaliates against the sound that dodging around the Shade produces. This emergent upsurge is accomplished using one inter-agent stimulus - no collective state machine is needed. Complex outcomes Simple rules (Fullerton emergent complexity principle).

Feedback State Justification (Librande)

The feedback states are in accordance to Librande principle of information transparency - the player should always know the agent state. ! and ? icons are conventions of stealth-games that indicate the level of attention without interrupting the game. The ripple VFX and heartbeat SFX of the Shade have been created with a gentle audio-visual enhancing, which befits its covert character - the intensity of feedback is related to the identity of the agent. The Sentinel proclaims itself in a noisy way; the Shade usurps noiselessly.